Numerical Methods for Elliptic Problems

Exercise Sheet 1

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Exercise 1 (Vector identities, integration-by-parts)

(a)

$$\operatorname{div}(\vec{b}u) = \sum_{i=1}^{d} \partial_i(ub_i) \overset{\text{PR}}{=} \sum_{i=1}^{d} ((\partial_i u)b_i + u\partial_i b_i) = \vec{b}\cdot\nabla u + u\operatorname{div}\vec{b}.$$

(b) Gauß:

$$\int_{\partial\Omega} u\vec{b}\cdot\vec{n}\,\mathrm{d}s = \int_\Omega \operatorname{div}(u\vec{b})\,\mathrm{d}x \overset{(a)}{=} \int_\Omega \vec{b}\cdot\nabla u\,\mathrm{d}x + \int_\Omega u\operatorname{div}\vec{b}\,\mathrm{d}x.$$

(c) In (b) choose $\vec{b} = \nabla v$, then

$$\int_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x = -\int_\Omega u\underbrace{\operatorname{div}(\nabla v)}_{\Delta v}\,\mathrm{d}x + \int_{\partial\Omega} u\underbrace{\nabla v\cdot\vec{n}}_{\partial_n v}\,\mathrm{d}s.$$

Exchange u, v and use symmetry of inner product:

$$\int_\Omega \nabla u\cdot\nabla v\,\mathrm{d}x = -\int_\Omega \Delta u\, v\,\mathrm{d}x + \int_{\partial\Omega} \partial_n u\, v\,\mathrm{d}s.$$

Exercise 2 (Fundamental lemma 1)

1. $\Leftarrow$ Clear.

$\Rightarrow$ Assume for the sake of contradiction there exists $x_0 \in \Omega$ with $f(x_0) \neq 0$. Without loss $f(x_0) > 0$. Due to the continuity of f, there exists $\delta > 0$ such that $f(x) > 0$ for $x \in B_\delta(x_0) \subset \Omega$ with δ sufficiently small. Now choose $r < \delta$ such that $\overline{B}_r(x_0) \subset \Omega$. It holds:

$$0 = \int_{B_\delta(x_0)} f(x)\,\mathrm{d}x \geq \int_{\overline{B}_r(x_0)} \underbrace{f(x)}_{>0}\,\mathrm{d}x > 0. \qquad \perp$$

2. $\Leftarrow$ $f = 0$ a.e. $\Rightarrow \int_\omega f\,\mathrm{d}x = 0$ (clear).

$\Rightarrow$ We show that the set $B = \{x \in \Omega : f(x) \neq 0\}$ has measure zero. For this choose a "subset on which f is strictly positive":

$$E_{n,R} = \left\{x \in \Omega \cap B_R(0) : f(x) \geq \frac{1}{n}\right\}.$$

Clearly for $x \in E_{n,R}$: $f(x) \geq \frac{1}{n} > 0$, for all n, R by definition. Also $E_{n,R}$ measurable (clear) and bounded due to "$\cap B_R(0)$" for $R < \infty$. Now,

$$0 \overset{\text{Ass.}}{=} \int_{E_{n,R}} \underbrace{f(x)}_{\geq 1/n}\,\mathrm{d}x \geq \underbrace{\frac{1}{n}}_{>0}\underbrace{|E_{n,R}|}_{\geq 0} \quad \Rightarrow \quad |E_{n,R}| = 0.$$

Analogously, for $E'_{n,R} = \{x \in \Omega \cap B_R(0) : f(x) \leq -\frac{1}{n}\}$ it follows that $f(x) < \frac{-1}{n} < 0$, $\mu(E'_{n,R}) = 0$. Now,

$$A_{n,R} = E_{n,R} \cup E'_{n,R}$$

and

$$B \subset \bigcup_{n,R\in\mathbb{N}} A_{n,R}, \qquad \mu\left(\bigcup_{n,R\in\mathbb{N}} A_{n,R}\right) = 0 \quad \Rightarrow \quad \mu(B) = 0.$$

Equivalently, $f = 0$ a.e.

Exercise 3 (Fundamental lemma, 2nd version)

$\Leftarrow$

$$f(x) = 0 \Rightarrow f(x)v(x) = 0 \quad \forall v \in C(\overline{\Omega}) \Rightarrow 0 = \int_\Omega fv \,\mathrm{d}x \quad \forall v.$$

$\Rightarrow$ Suppose $f(x_0) \neq 0$ for $x_0 \in \Omega$. W.l.o.g. $f(x_0) > 0$. Continuity: $\overline{B}_r(x_0) \subset \Omega$ with $f(x) > 0$ for all $x \in \overline{B}_r(x_0)$. Define

$$v(x) = \max\left\{1 - \frac{|x - x_0|}{r}, 0\right\}, \qquad x \in \overline{\Omega}.$$

Then $v \geq 0$, $v(x_0) = 1$, $v(x) = 0$ outside $B_r(x_0)$ and v is continuous. In particular $v|_{\partial\Omega} = 0$, so v is admissible. Furthermore

$$f(x)v(x) > 0 \quad \text{on } B_r(x_0), \qquad f(x)v(x) = 0 \quad \text{outside } B_r(x_0).$$

Thus,

$$\int_\Omega f(x)v(x)\,\mathrm{d}x = \int_{B_r(x_0)} \underbrace{f(x)v(x)}_{>0}\,\mathrm{d}x > 0. \qquad \perp$$

L^2 **case:** $\Leftarrow$ Clearly $f = 0$ a.e. $\Rightarrow \int_\Omega fv\,\mathrm{d}x = 0$ for all $v \in L^2(\Omega)$. As Ω is bounded, if $f \in C_c^\infty(\Omega)$ then $f \in L^2(\Omega)$. Thus $C_c^\infty(\Omega) \subset L^2(\Omega)$ and thus the leftmost implication "$\Leftarrow$" holds.

1. $\Rightarrow$ Suppose $\int_\Omega f\varphi\,\mathrm{d}x = 0$ for all $\varphi \in C_c^\infty(\Omega)$. Let $v \in L^2(\Omega)$ arbitrary but fixed. We show $\int_\Omega fv\,\mathrm{d}x = 0$. Due to density of $C_c^\infty(\Omega)$ in $L^2(\Omega)$ there exists a sequence $(\varphi_n) \subset C_c^\infty$:

$$\|\varphi_n - v\|_{L^2(\Omega)} \to 0.$$

Due to C–S-inequality:

$$|\langle f, v - \varphi_n\rangle|_{L^2} \leq \|f\|_{L^2} \underbrace{\|v - \varphi_n\|_{L^2}}_{\to 0} \to 0.$$

Due to assumption: $\langle f, \varphi_n\rangle = 0$ for all n, as $\varphi_n \in C_c^\infty$.

$$\begin{aligned}\langle f, v\rangle &= \langle f, v - \varphi_n + \varphi_n\rangle \\ &= \underbrace{\langle f, v - \varphi_n\rangle}_{\leq|\cdot|} + \underbrace{\langle f, \varphi_n\rangle}_{0} = \langle f, v - \varphi_n\rangle \to 0.\end{aligned}$$

Hence, it must also hold that $\langle f, v\rangle = 0$.

2. $\Rightarrow$ Suppose $\int_\Omega fv\,\mathrm{d}x = 0$ for all $v \in L^2(\Omega)$. As $f \in L^2$, in particular $v = f$:

$$0 = \int_\Omega f^2\,\mathrm{d}x = \|f\|_{L^2}^2 \quad \Rightarrow \quad f = 0.$$

Exercise 4

(a) We compute the first variation

$$\begin{aligned}\delta J(u)[v] &= \lim_{t\to 0}\frac{1}{t}\big[J(u+tv)-J(u)\big]\\ &= \lim_{t\to 0}\frac{1}{2t}\Big[\int_\Omega a|\nabla(u+tv)|^2\,\mathrm{d}x - \int_\Omega fu\,\mathrm{d}x - 2t\int_\Omega fv\,\mathrm{d}x\\ &\qquad - \int_\Omega a|\nabla u|^2\,\mathrm{d}x + \int_\Omega fu\,\mathrm{d}x\Big].\end{aligned}$$

$$\nabla(u+tv) = \nabla u + t\nabla v, \qquad |\nabla u + t\nabla v|^2 = |\nabla u|^2 + 2t\nabla u\cdot\nabla v + t^2|\nabla v|^2.$$

$$\begin{aligned}\delta J(u)[v] &= \lim_{t\to 0}\frac{1}{2t}\Big[\int_\Omega a|\nabla u|^2\,\mathrm{d}x + 2t\int_\Omega a\nabla u\cdot\nabla v\,\mathrm{d}x + t^2\int_\Omega a|\nabla v|^2\,\mathrm{d}x\\ &\qquad - 2t\int_\Omega fv\,\mathrm{d}x - \int_\Omega a|\nabla u|^2\,\mathrm{d}x\Big]\\ &= \int_\Omega a\nabla u\cdot\nabla v\,\mathrm{d}x - \int_\Omega fv\,\mathrm{d}x + \underbrace{\frac{1}{2}\lim_{t\to 0} t\int_\Omega a|\nabla v|^2\,\mathrm{d}x}_{0}.\end{aligned}$$

$$\int_\Omega a|\nabla v|^2\,\mathrm{d}x \le \|a\|_{L^\infty(\Omega)}\|\nabla v\|_{L^2}^2, \qquad \text{because we can choose } v\in H_0^1(\Omega).$$

Setting the first variation to zero we obtain:

$$\int_\Omega a\nabla u\cdot\nabla v\,\mathrm{d}x = \int_\Omega fv\,\mathrm{d}x \qquad \forall v\in H_0^1.$$

Comparison to min surface:

$$\int_\Omega \frac{1}{\sqrt{1+|\nabla u|^2}}\nabla u\cdot\nabla v\,\mathrm{d}x = 0, \qquad \forall v:\ v|_{\partial\Omega}=0.$$

Here, we are linear in ∇u, ∇v: $a\nabla u\cdot\nabla v$. MS is non-linear: $\frac{1}{\sqrt{1+|\nabla u|^2}}\nabla u\cdot\nabla v$. However, we have a RHS $\int_\Omega fv\,\mathrm{d}x$.

(b) Using (a) from Ex 1 and restricting to $v\in C_c^\infty\subset H_0^1$:

$$\operatorname{div}\big((a\nabla u)v\big) = (a\nabla u)\cdot\nabla v + v\operatorname{div}(a\nabla u), \qquad \forall v\in C_c^\infty.$$

$$\int_\Omega \operatorname{div}\big((a\nabla u)v\big)\,\mathrm{d}x = \underbrace{\int_\Omega a\nabla u\cdot\nabla v\,\mathrm{d}x}_{\int_\Omega fv\,\mathrm{d}x\ \text{(Teil a)}} + \int_\Omega v\operatorname{div}(a\nabla u)\,\mathrm{d}x.$$

Gauß:

$$\int_{\partial\Omega}\big((a\nabla u)v\big)\cdot\vec{n}\,\mathrm{d}s = 0 \qquad (v=0 \text{ due to } v\in C_c^\infty).$$

$$\begin{aligned}0 &= \int_\Omega fv\,\mathrm{d}x + \int_\Omega v\operatorname{div}(a\nabla u)\,\mathrm{d}x\\ &= \int_\Omega v\big(f+\operatorname{div}(a\nabla u)\big)\,\mathrm{d}x \qquad \forall v\in C_c^\infty.\end{aligned}$$

By 3 we obtain:

$$f+\operatorname{div}(a\nabla u)=0 \quad\Leftrightarrow\quad -\operatorname{div}(a\nabla u) = f \quad \text{in } \Omega,$$

and boundary condition

$$u = 0 \quad \text{on } \partial\Omega,$$

from the definition of the minimization problem.

Exercise 5

Let $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be a classical solution of

$$-\Delta u = 0 \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega.$$

Choose $w = u - u_g$, then it is admissible:

$$-\Delta w = \underbrace{-\Delta u}_{0} + \Delta u_g = \Delta u_g = f \quad \text{in } \Omega.$$

$$w = \underbrace{u}_{g} - \underbrace{u_g}_{g} = 0.$$

Furthermore,

$$u = u_g + w = u_g + u - u_g = u.$$

Exercise 6

$$-\operatorname{div}(-\vec{b}u) = \operatorname{div}(\vec{b}u) = \vec{b} \cdot \nabla u + u \operatorname{div}(\vec{b}).$$

Thus, the equation is:

$$\vec{b} \cdot \nabla u + u \operatorname{div}(\vec{b}) + cu = f.$$

$$\underbrace{\vec{b}}_{=\vec{b}'} \cdot \nabla u + \underbrace{\left(\operatorname{div}(\vec{b}) + c\right)}_{=c'} u = \underbrace{f}_{=f'}.$$

$$\vec{b}' \cdot \nabla u = 0.$$

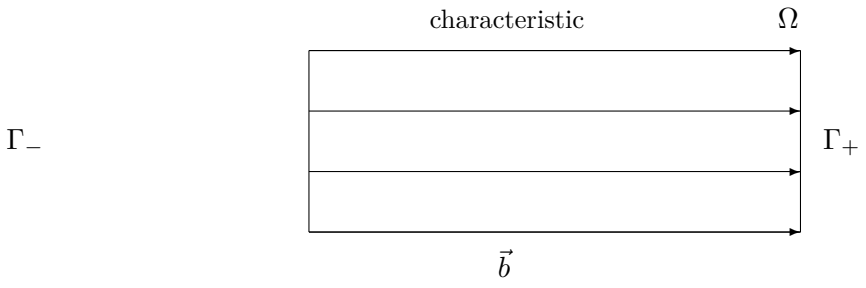
Take characteristic segment $x(t) = x_0 + t\vec{b}'$. Note that due to the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}t} u(x(t)) = \nabla u \cdot x'(t) = \nabla u \cdot \vec{b}',$$

equivalently

$$\frac{\mathrm{d}}{\mathrm{d}t} u(x(t)) = 0.$$

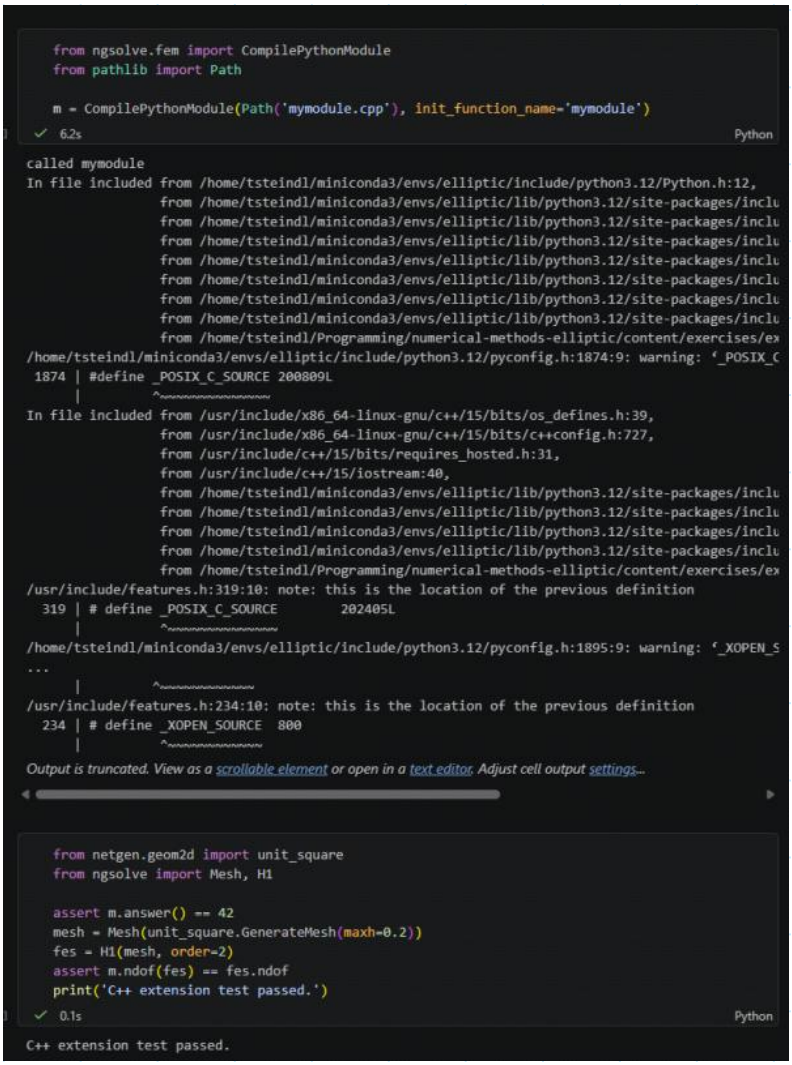
This means along segments in direction $\vec{b}'$, the solution u is constant.



$$\Gamma_- = \{x \in \partial\Omega : \vec{b}' \cdot \vec{n}(x) < 0\}.$$

Since u is constant along horizontal curves, its value is determined by the value that enters the segment (inflow). Both the interior and outflow are determined by this.

Exercise 7



```
from ngsolve.fem import CompilePythonModule
from pathlib import Path

m = CompilePythonModule(Path('mymodule.cpp'), init_function_name='mymodule')
```

✓ 6.2s Python

```
called mymodule
In file included from /home/tsteindl/miniconda3/envs/elliptic/include/python3.12/Python.h:12,
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
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                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/Programming/numerical-methods-elliptic/content/exercises/ex
/home/tsteindl/miniconda3/envs/elliptic/include/python3.12/pyconfig.h:1874:9: warning: ‘_POSIX_C
 1874 | #define _POSIX_C_SOURCE 200809L
      |         ^~~~~~~~~~~~~~~~~~~~~~~
In file included from /usr/include/x86_64-linux-gnu/c++/15/bits/os_defines.h:39,
                 from /usr/include/x86_64-linux-gnu/c++/15/bits/c++config.h:727,
                 from /usr/include/c++/15/bits/requires_hosted.h:31,
                 from /usr/include/c++/15/iostream:40,
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/miniconda3/envs/elliptic/lib/python3.12/site-packages/inclu
                 from /home/tsteindl/Programming/numerical-methods-elliptic/content/exercises/ex
/usr/include/features.h:319:10: note: this is the location of the previous definition
  319 | # define _POSIX_C_SOURCE        202405L
      |          ^~~~~~~~~~~~~~~~~~~~~~
/home/tsteindl/miniconda3/envs/elliptic/include/python3.12/pyconfig.h:1895:9: warning: ‘_XOPEN_S
...
      |         ^~~~~~~~~~~~~~~
/usr/include/features.h:234:10: note: this is the location of the previous definition
  234 | # define _XOPEN_SOURCE  800
      |          ^~~~~~~~~~~~~~~
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```

```
from netgen.geom2d import unit_square
from ngsolve import Mesh, H1

assert m.answer() == 42
mesh = Mesh(unit_square.GenerateMesh(maxh=0.2))
fes = H1(mesh, order=2)
assert m.ndof(fes) == fes.ndof
print('C++ extension test passed.')
```

✓ 0.1s Python

```
C++ extension test passed.
```